

A Machine Learning Model for the Prediction of Hessian Matrices based on Semiempirical Electronic Structure Theory

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Introduction

- The Hessian matrix shall be computed with a machine learning (ML) model
- Hessian matrices are computational costly and thus are a valuable target for accelerated computation.
- The ML model is an ensemble based decision tree method and uses atomistic features derived from GFN2-xTB^[1]

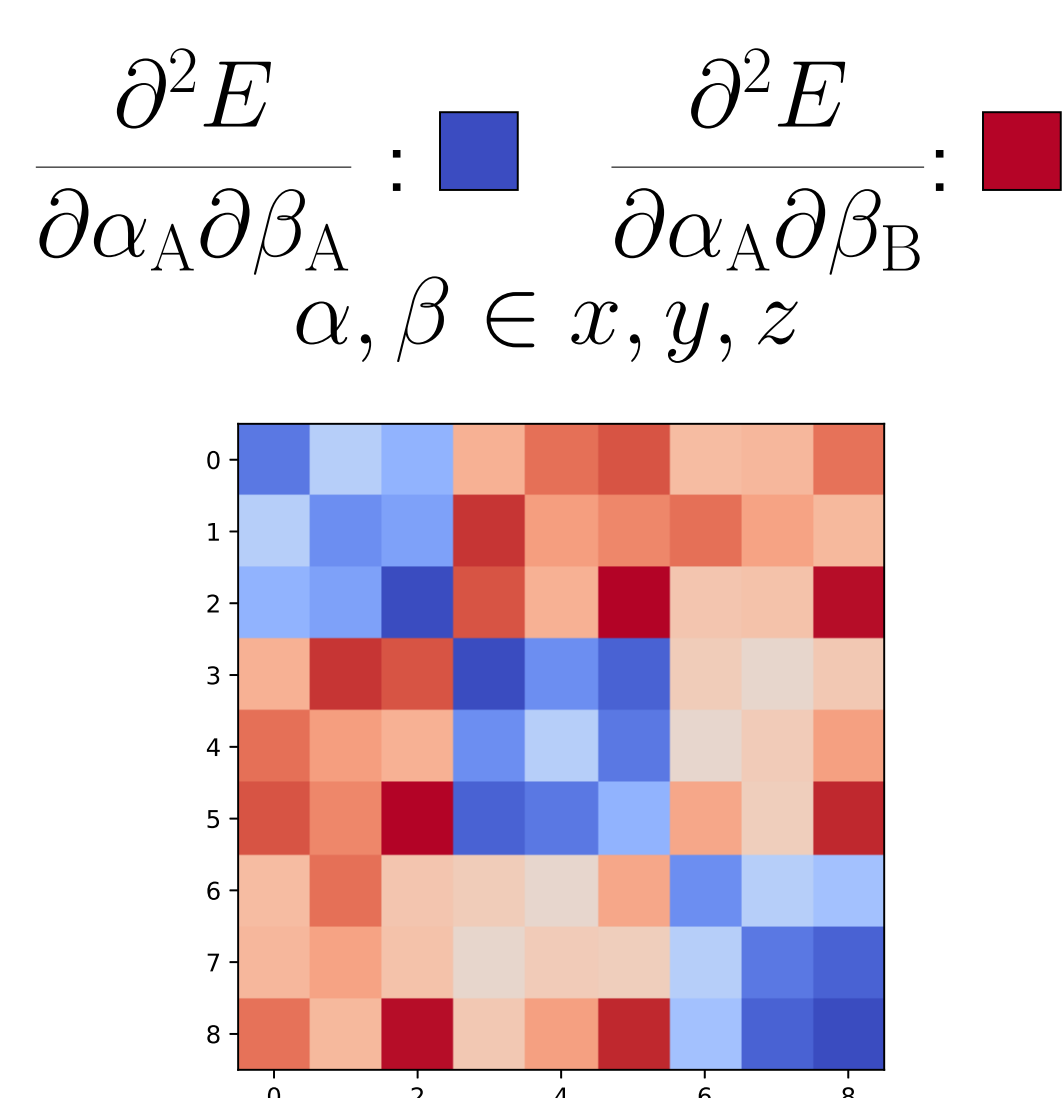
Overview of the ML Model

Employed Features

- Atomically partitioned features dependent on the geometry, electron density, and individual contributions to the total energy.
- Mulliken partitioning has been used for the generation of atomistic features.

The Hessian Matrix

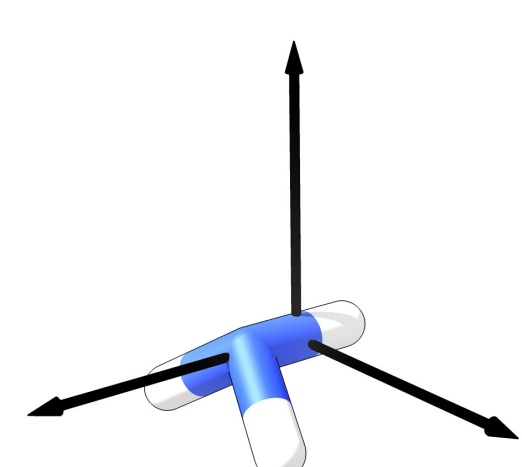
- The elements of the Hessian are second derivatives either with respect to the coordinates of a single atom or of two atoms.
- For the machine learning (ML) model, we distinguish between the diagonal blocks (blue, same atoms) and off-diagonal blocks (red, different atoms).



Connecting Features and Targets

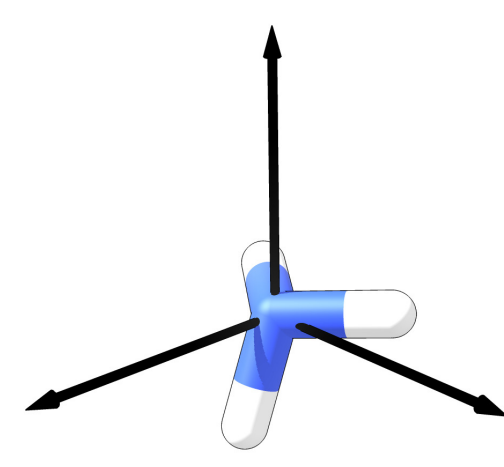
- The feature vector is either built from the quantities of one or two atoms.
- For ML modeling, a fixed feature vector length is necessary, hence two separate models are used which rely on ensemble-based decision tree methods.
- The features and targets are provided for each 3 x 3 block in a model-specific orientation.

Model ■ : The origin of the coordinate system is shifted to the center of two atoms and the atoms are aligned to the z-axis.



- Predicts 9 unique Hessian matrix elements
- Uses features derived from atom A (q_A) and B (q_B) as well as connections of q_A and q_B ($q_A - q_B$; $q_A + q_B$; $q_A \cdot q_B$)

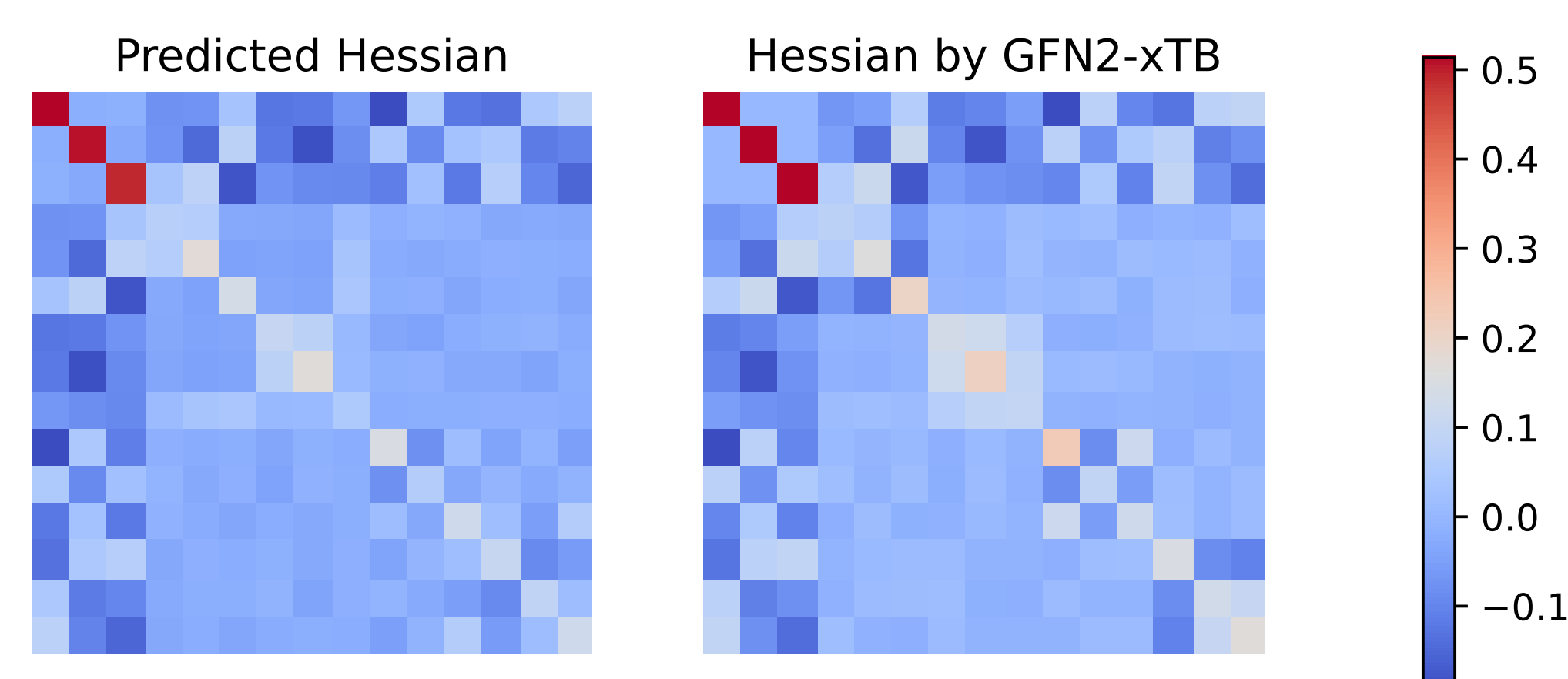
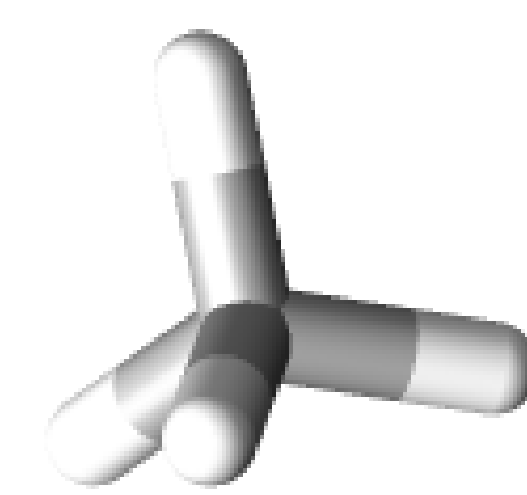
Model ■ : Here, the origin of the coordinate system is shifted to the atom.



- Predicts 6 unique Hessian matrix elements
- Only uses features derived from atom A (q_A)

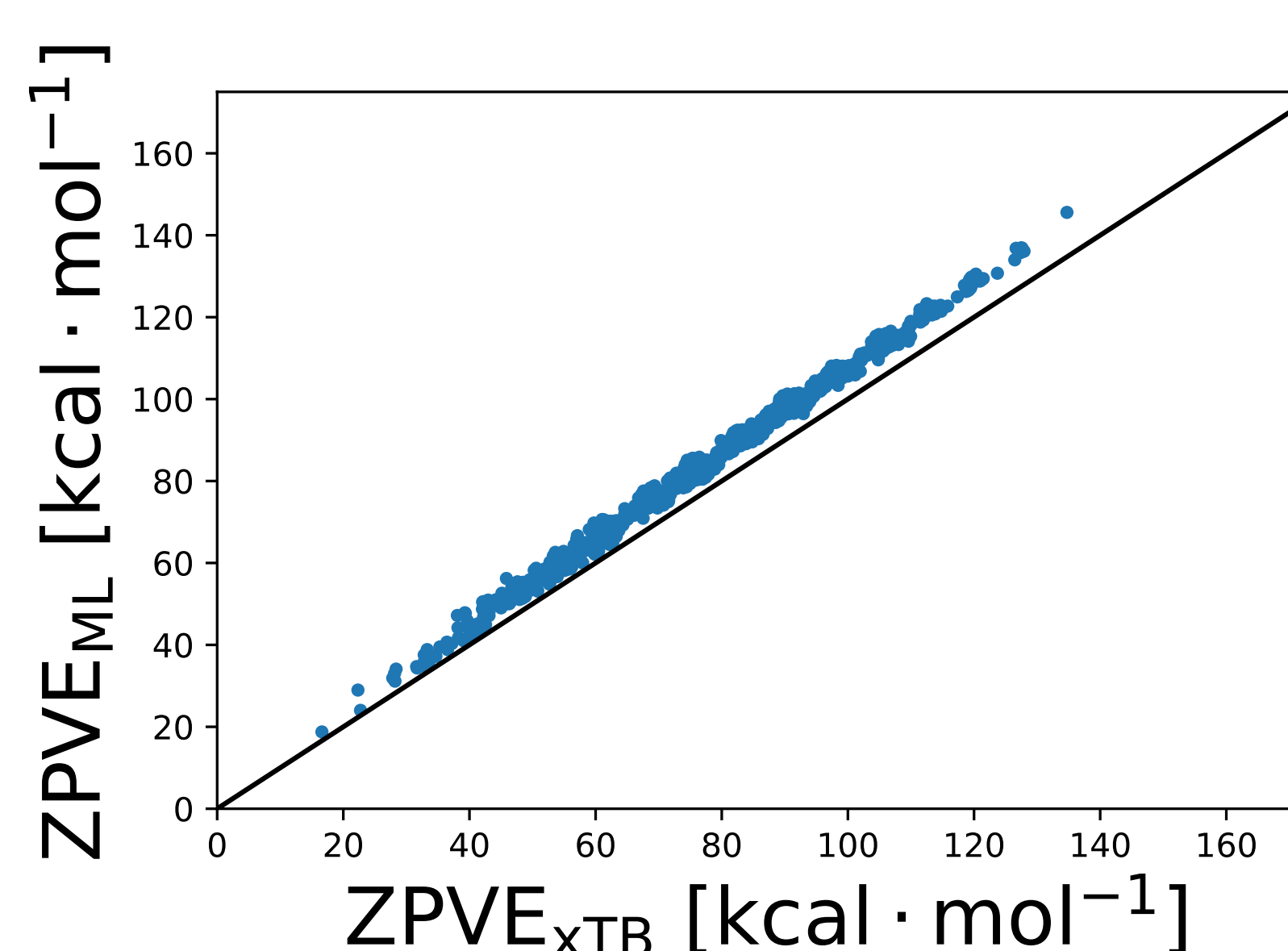
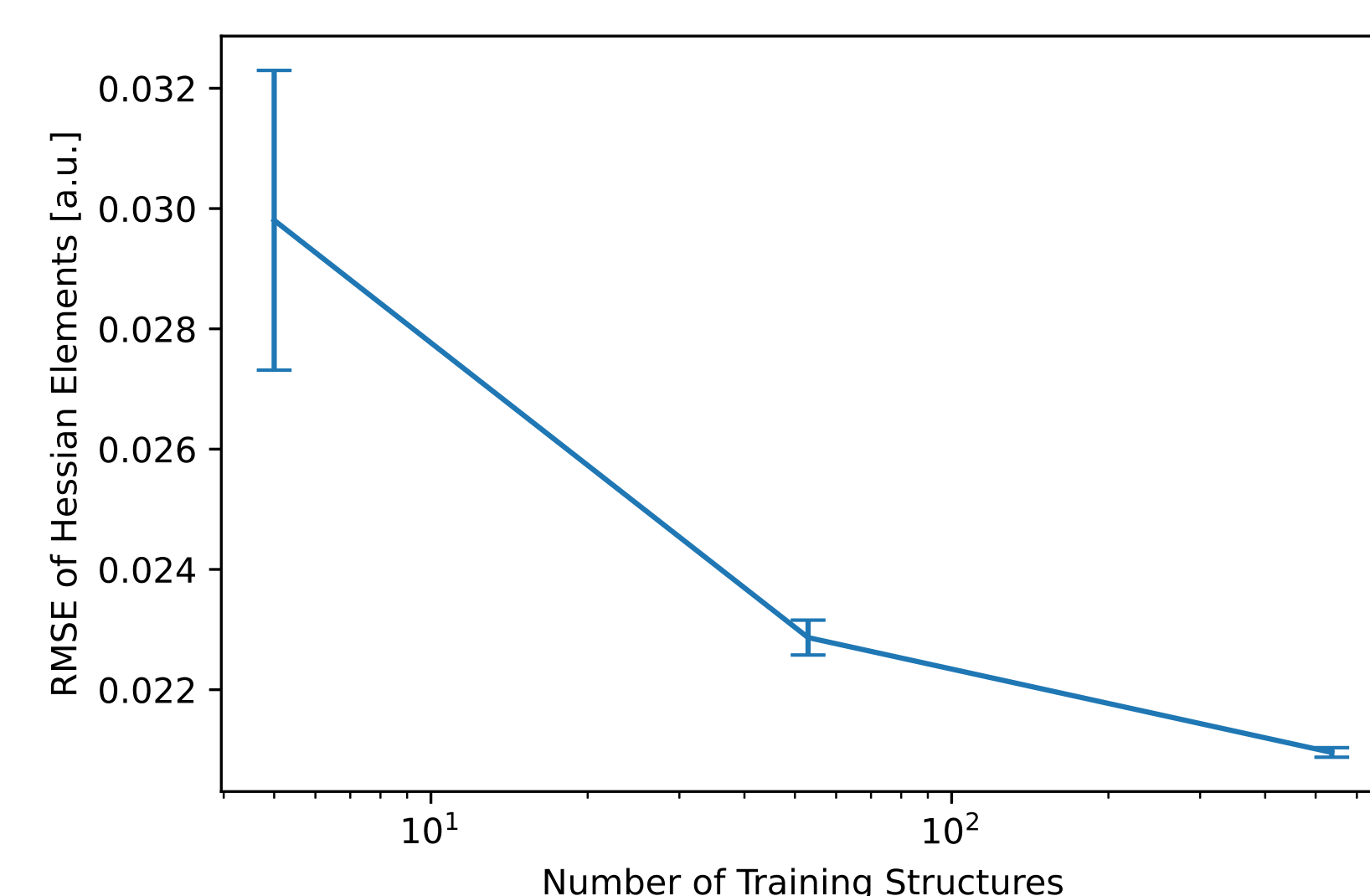
Comparison of predicted and GFN2-xTB Hessian

- Example: Hessian matrix of methane
- Overall good resemblance of the predicted Hessian compared to the computed GFN2-xTB Hessian.



Predicted ZPVE of GBD7 Set

- Learning plot is computed using the GDB7^[2] set, which consists of 7164 structures.
- The test set was fixed to 25 % of the size of the GBD7 set.
- The training set was increased up to 15% of the full set.
- Test and Training set are randomly selected.



- Predicted zero point vibrational energy (ZPVE) for the test set with 15 % training set.
- Low scattering of the data with a small systematic overestimation of the ZPVE.
- MD:
- RMSD: XZ
- rel. RMSD: XY %

Conclusion and Future Perspective

- An ML model for the prediction of Hessian matrix elements based on atomistic features from GFN2-xTB electronic structure was developed.
- This enables the accelerated approximate computation of Hessian matrices, which can be embedded in efficient computational workflows.
- Feature enhancement for improved prediction.
- Prediction of Hessian matrices obtained at a higher level of theory, e.g. from DFT

References

- [1] C. Bannwarth, S. Ehlert, S. Grimme, *Journal of Chemical Theory and Computation* **2019**, 15, 1652.
- [2] M. Rupp, A. Tkatchenko, K.-R. Müller, O. A. von Lilienfeld, *Physical Review Letters* **2012**, 108.

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